



Housing Recovery and Reconstruction Platform-Nepal (HRRP)

Orientation Program for *Partner organizations*

on

Dry Stone Masonry Construction

Organized by:

National Reconstruction Authority(NRA)

MOUD/District Project Implementation Unit (DLPIU)

Housing Recovery and Reconstruction Platform-Nepal (HRRP)

Why Dry Stone Masonry ?



- This Type of Construction Common in Northern Area of Nepal
- In Northern Parts People don't Prefer Mud mortar
- it is difficult to transport cement and steel since these sites are very remote and have no road access
- Many Dry Stone Masonry Houses are already under Construction

Dry Stone Building Practices



- Extensive longitudinal walls and open front



Dry Stone Building Practices



- Stand alone, most with open fronts



Dry Stone Building Practices



- Dry Stone masonry or Stone in mud buildings with heavy Stone roofs
- Presence of corner stones



Dry Stone Building Practices



- No other Earthquake Resisting Element



Dry Stone Building Practices



- Steep Gable



Dry Stone Building Practices



- No proper connection of ridge beam, rafter and bottom chord
- No proper connection of roof elements with wall



Dry Stone Building Practices



- Dry Stone masonry with light CGI sheets
- No earthquake resisting elements
- Presence of corner stones
- Use Gable wall



New Construction after Earthquake

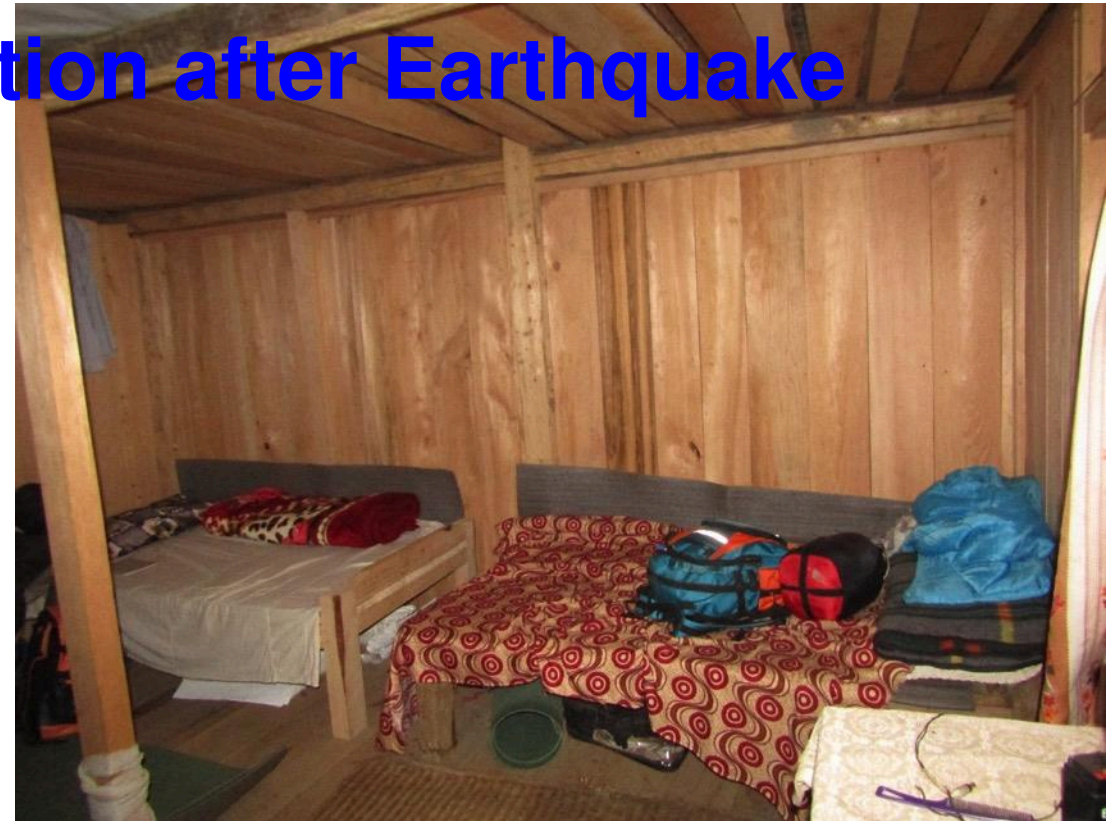
Dry Stone Building Practices



- Timber walls at the front part and inner side of all stone walls



New Construction after Earthquake



Many Buildings!

Dry Stone Construction-Deficiencies



- Gable End Wall Damage
- Heavy masonry gable walls are unsupported and subject to lateral loads from roof.



Dry Stone Construction-Deficiencies



- Short Wall Out-Of-Plane
- Weak attachment between end walls and diaphragms.
- Damage to and subsequent collapse of end walls due to out-of-plane demands.
- Wooden slab diaphragm unable to transfer Earthquake force Properly.



Dry Stone Construction-Deficiencies



- Wall Delamination in both directions
- The thick walls ($\geq 45\text{cm}$) constructed of interior and exterior facing stones. Required bonding elements are sometimes missing or inadequate.
- Wall delamination occurs, can lead to subsequent collapse.



Type Design On Dry Stone Masonry

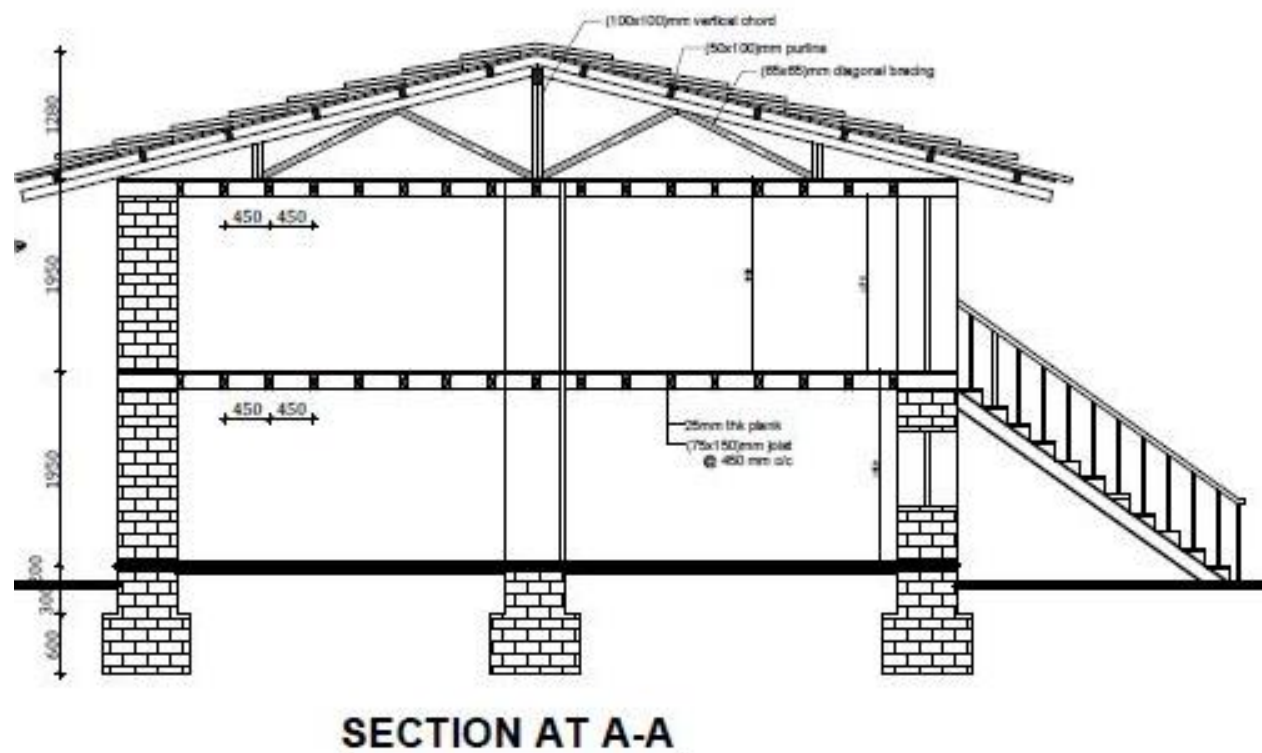
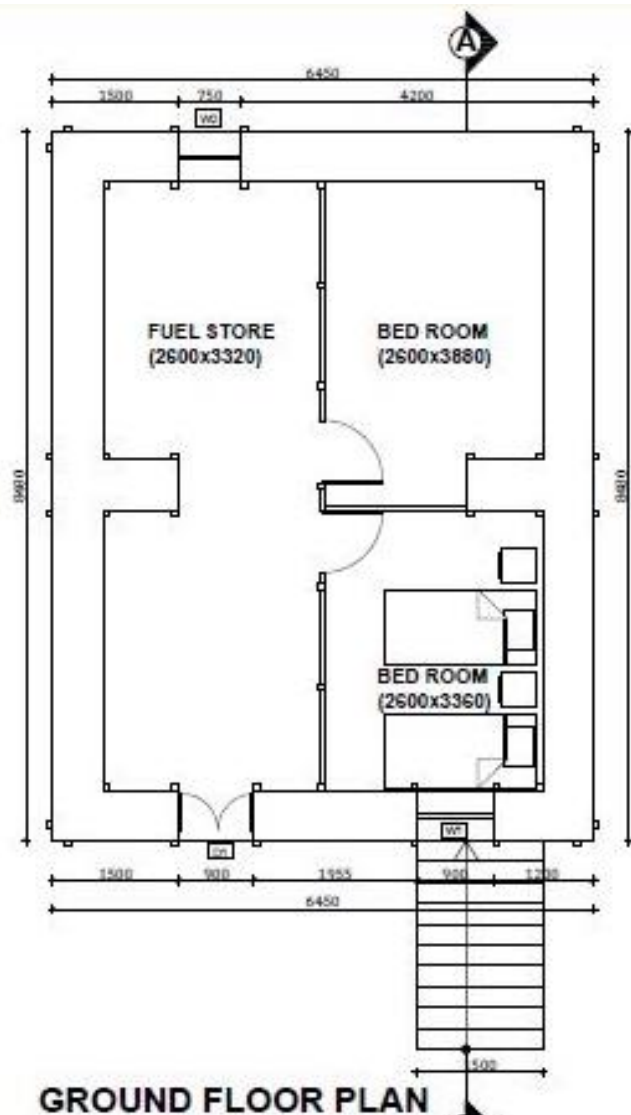


- Dressed Stone Masonry
- Random Rubble Masonry with GI Wire Mesh

Dressed Stone Masonry



- No. of Storey : 2
- Flexible Floor & Roof
- Plinth Area: 588 sq.ft

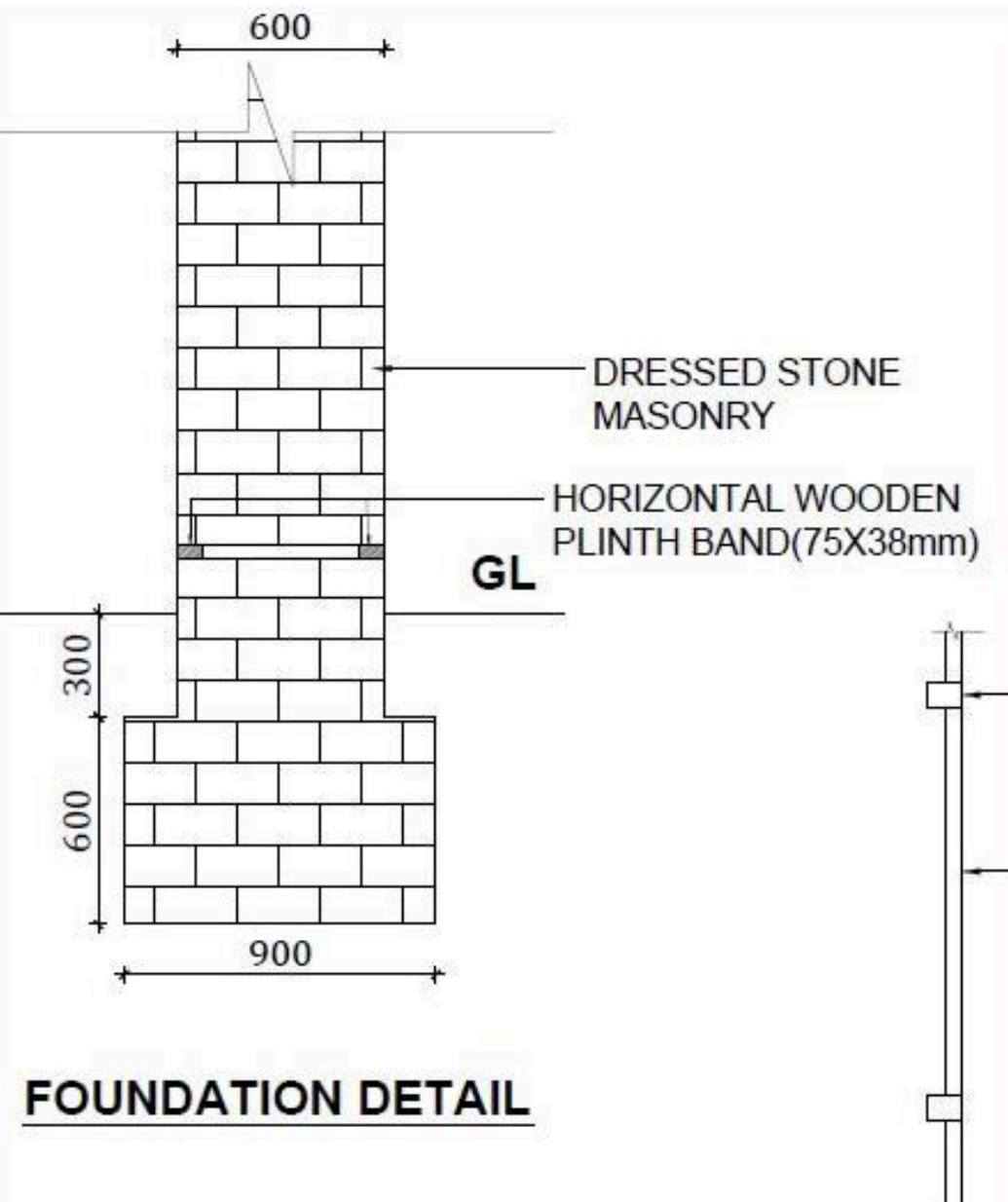


Dressed Stone Masonry

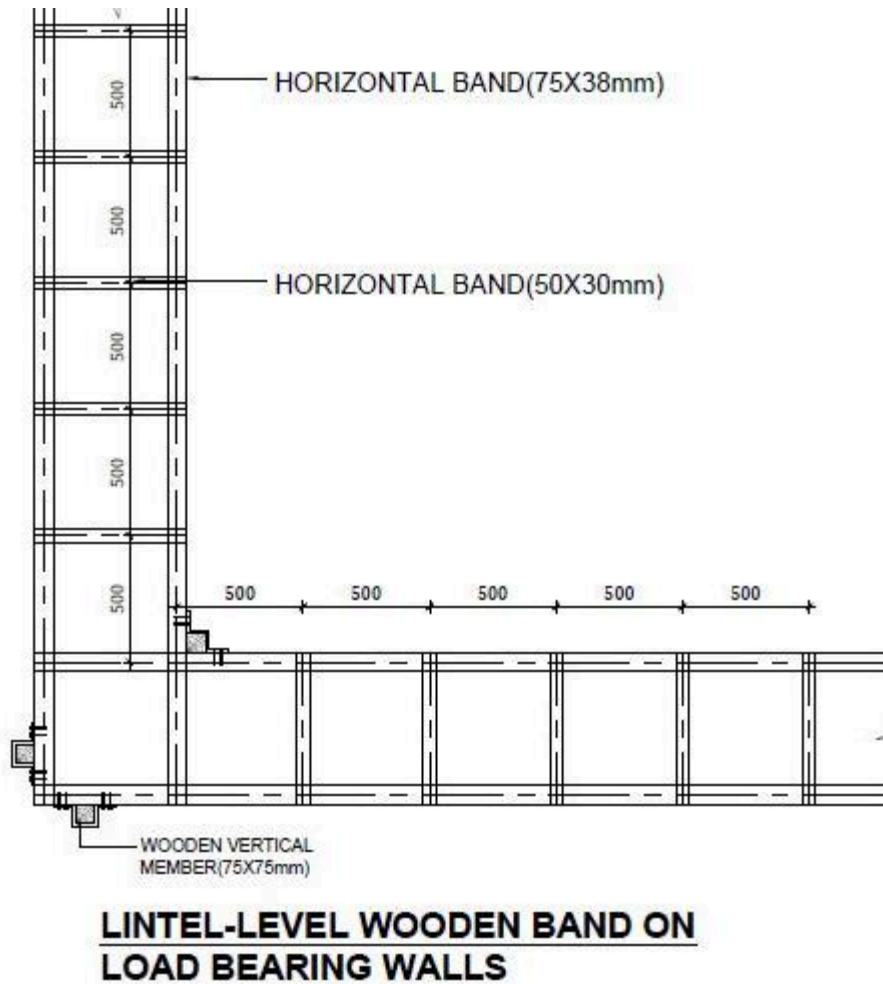


Foundation

- Width: 900 mm
- Depth : 900 mm



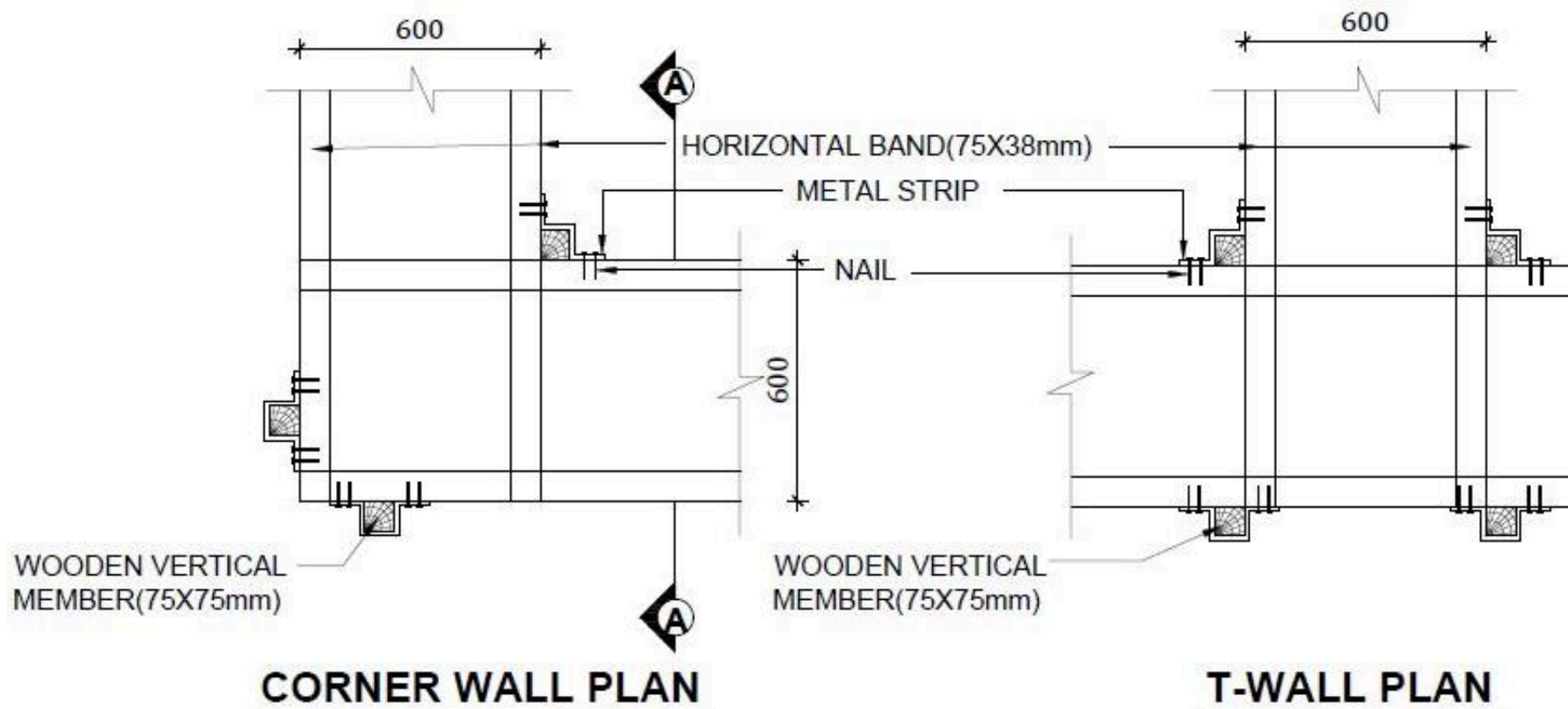
Dressed Stone Masonry



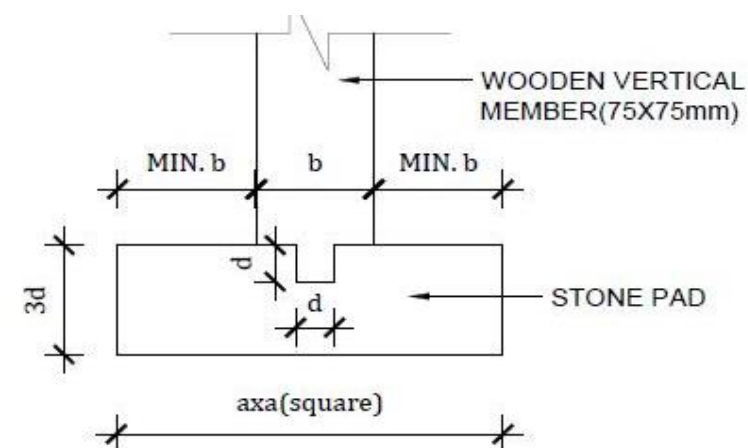
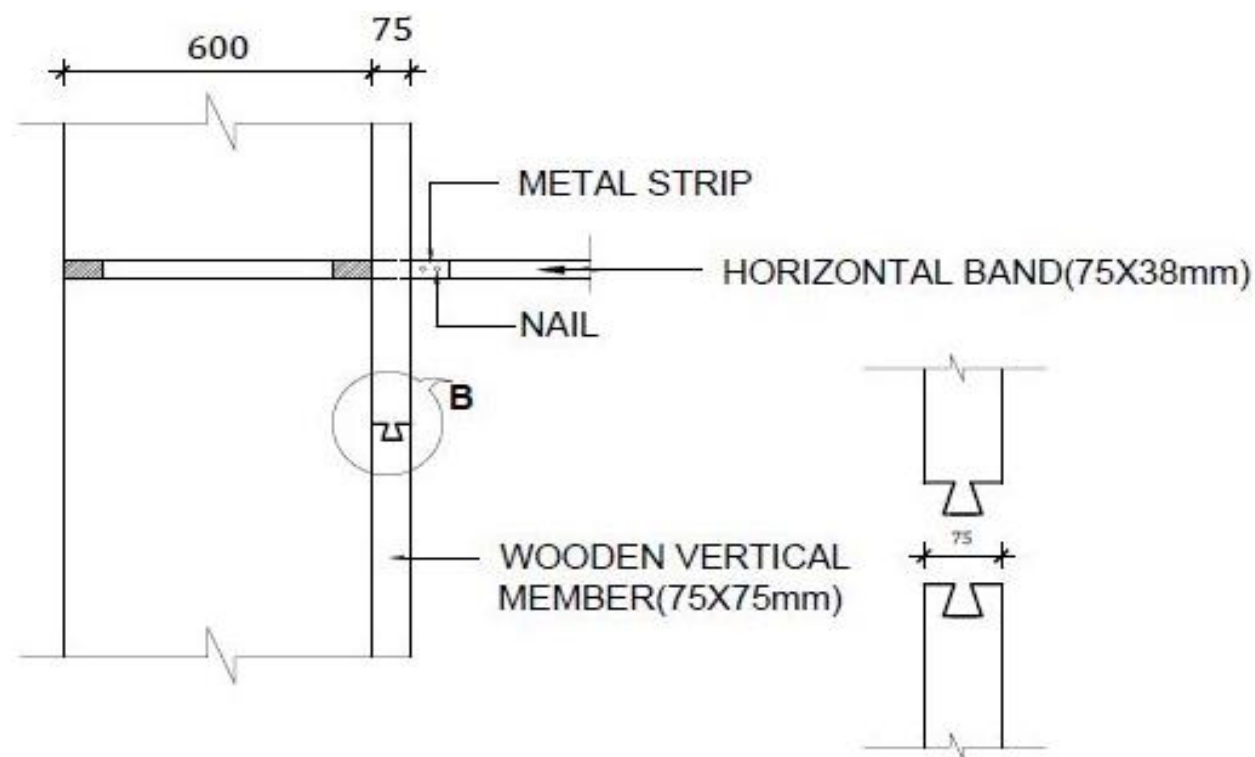
- Provide Horizontal Band at Plinth, Sill , Lintel & Roof level
- Well Seasoned Locally available hardwood can be used

Dressed Stone Masonry

- Wall Thickness : 600mm
- Spacing of through stone should not be more than 1200 mm horizontally & 600 mm Vertically
- Provide Vertical Member at Corner of Wall



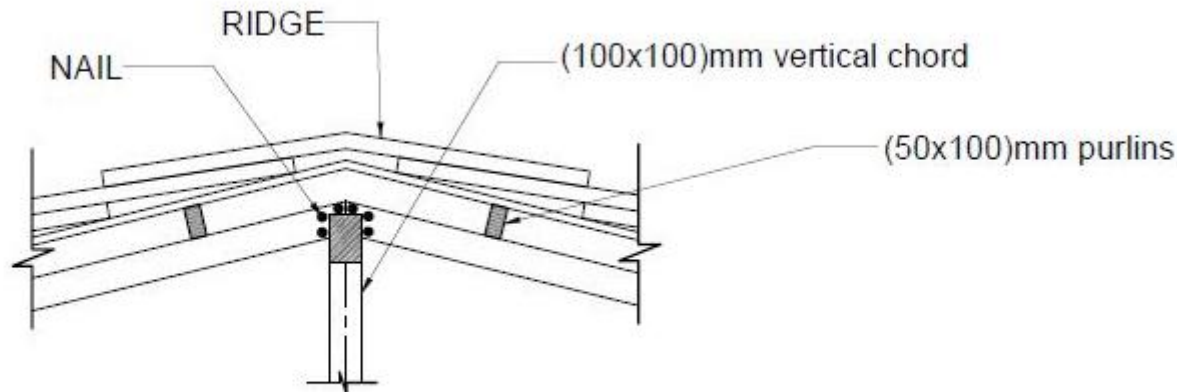
Vertical Member Detailing



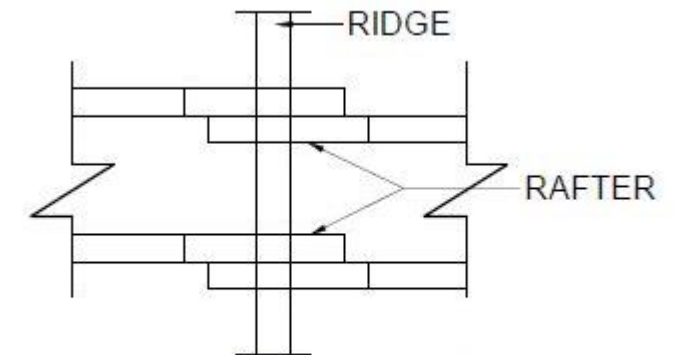
DETAIL OF FIXING POST ON BASE PAD

a= 3b OR GREATER
b=SIZE OF POST
d=b/3 OR 50mm

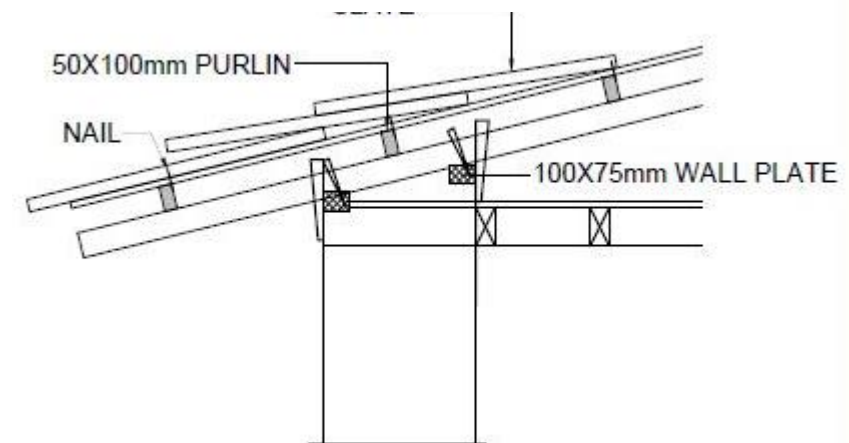
Roof Detailing



DETAIL OF RAFTER JOINT AT RIDGE

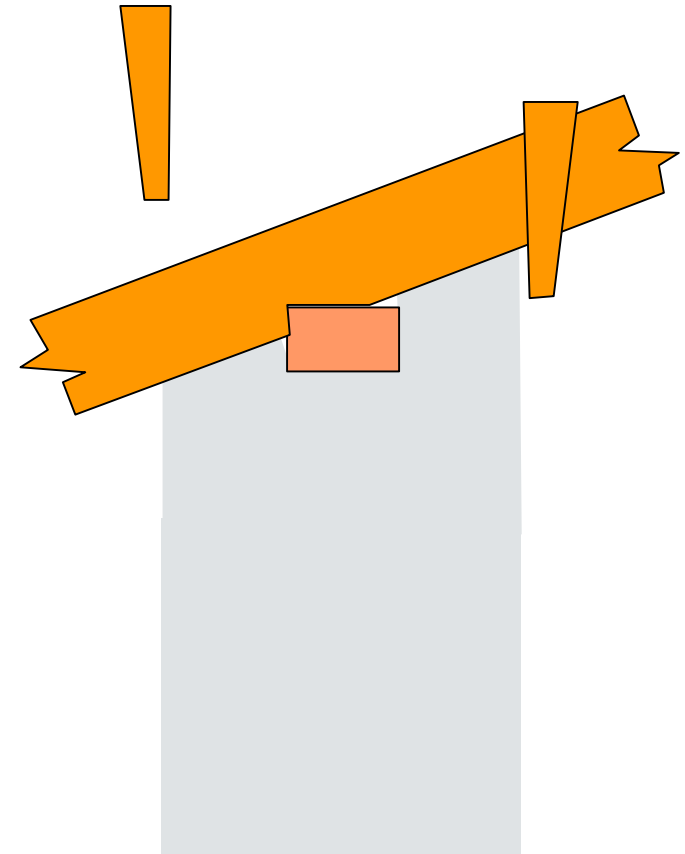
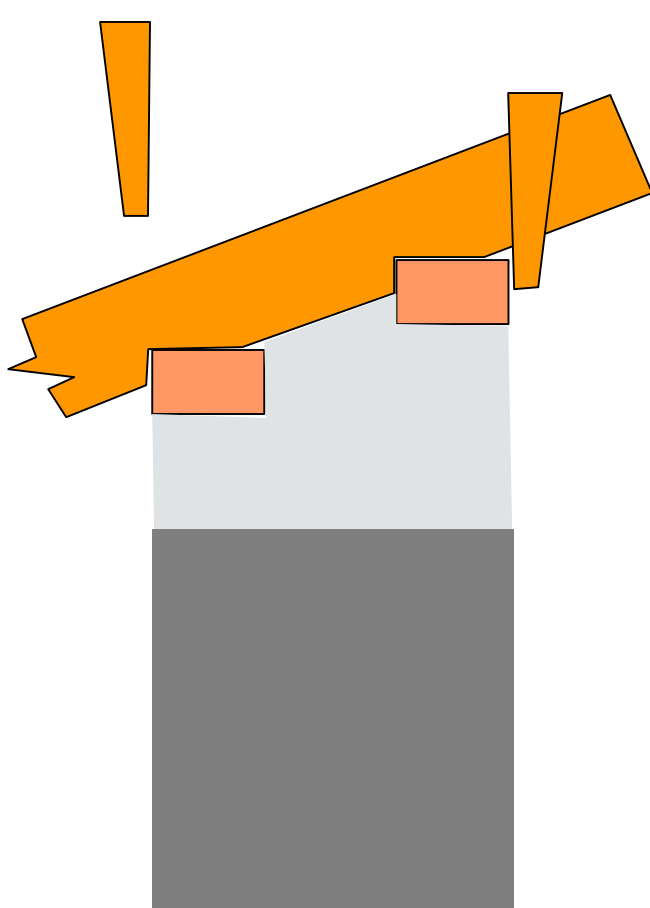


DETAIL PLAN AT RIDGE



In timber connection it need give Special attention in Joint !

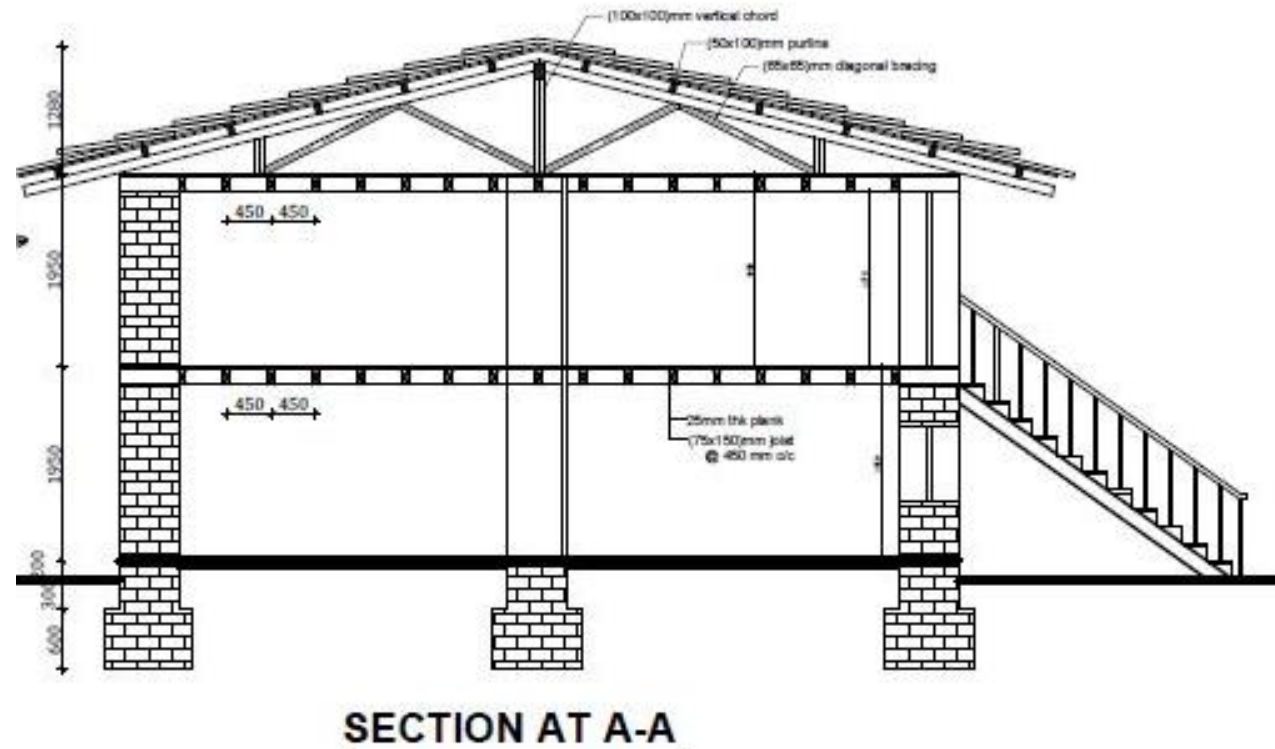
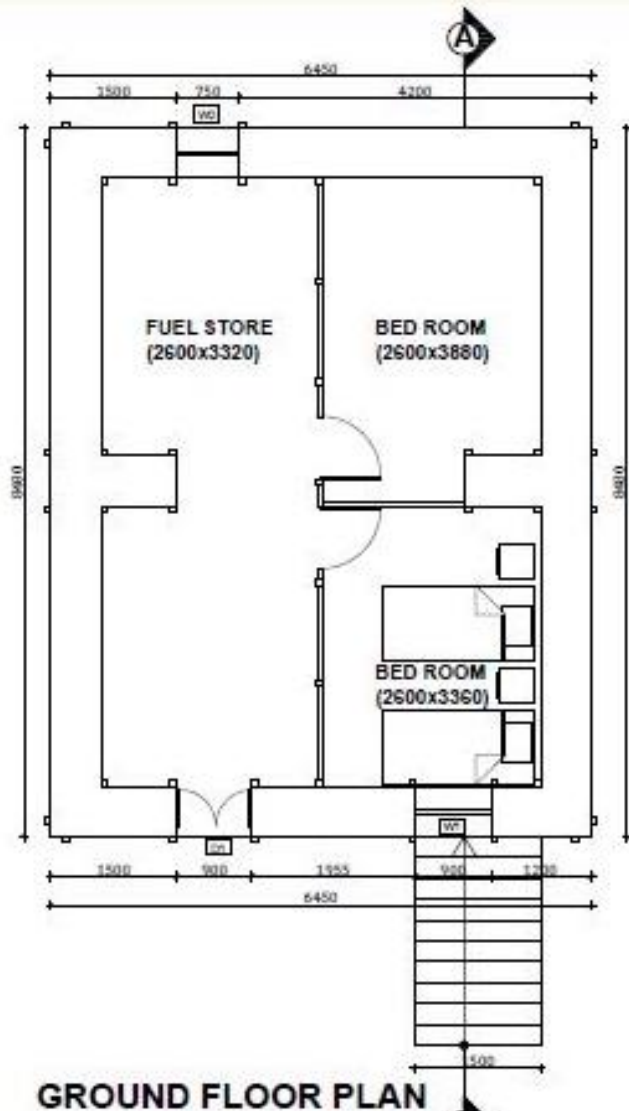
Wooden Key



Random Rubble Masonry with GI wire Mesh



- No. of Storey : 2
- Flexible Floor & Roof
- Plinth Area: 588 sq.ft

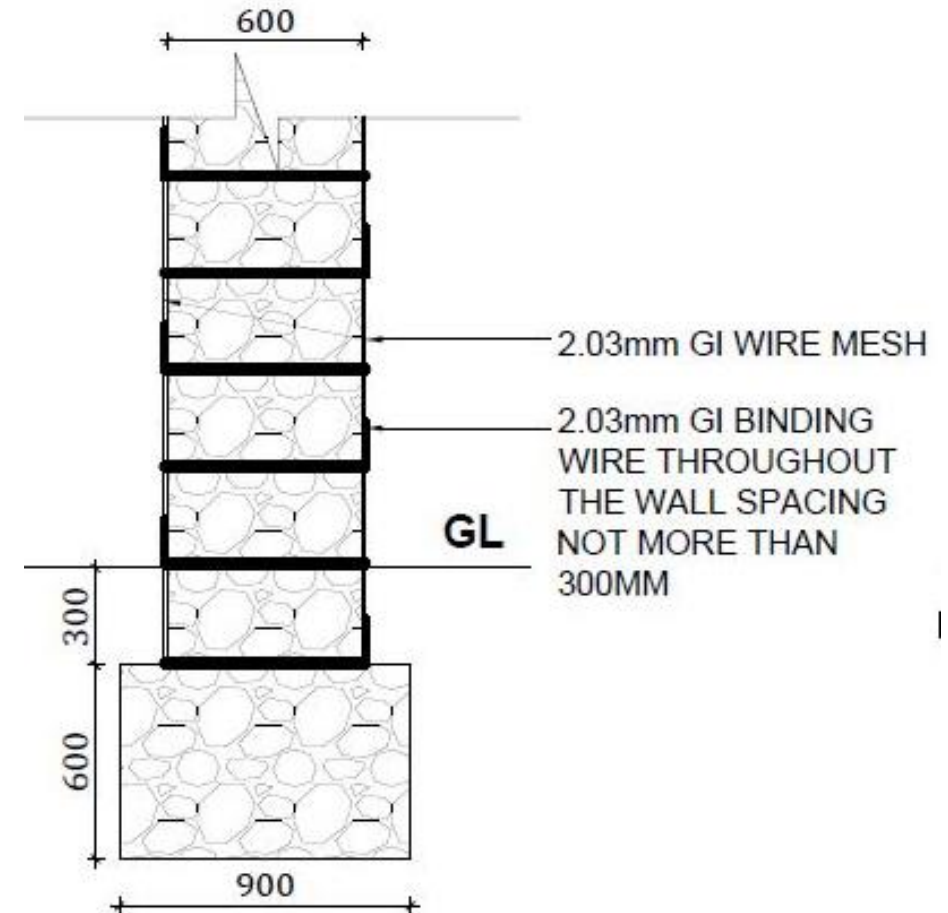


Random Rubble Masonry with GI wire Mesh



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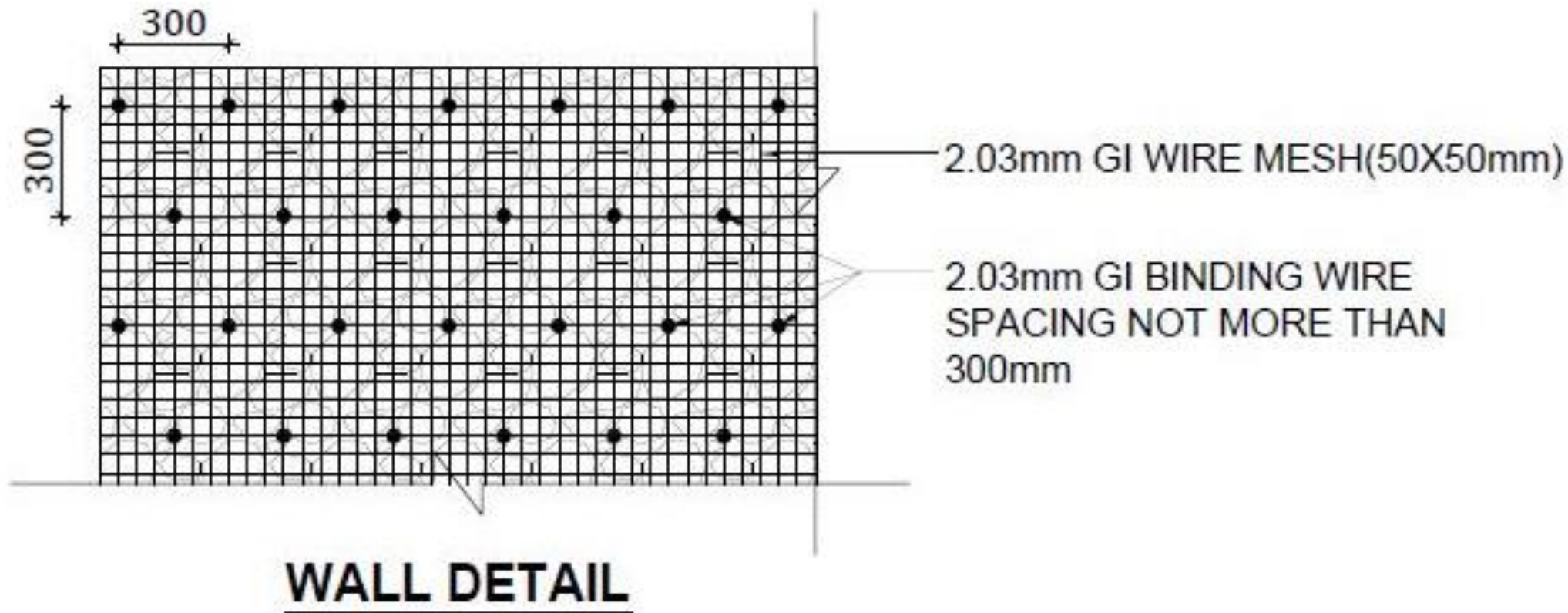


FOUNDATION DETAIL

Random Rubble Masonry with GI wire Mesh



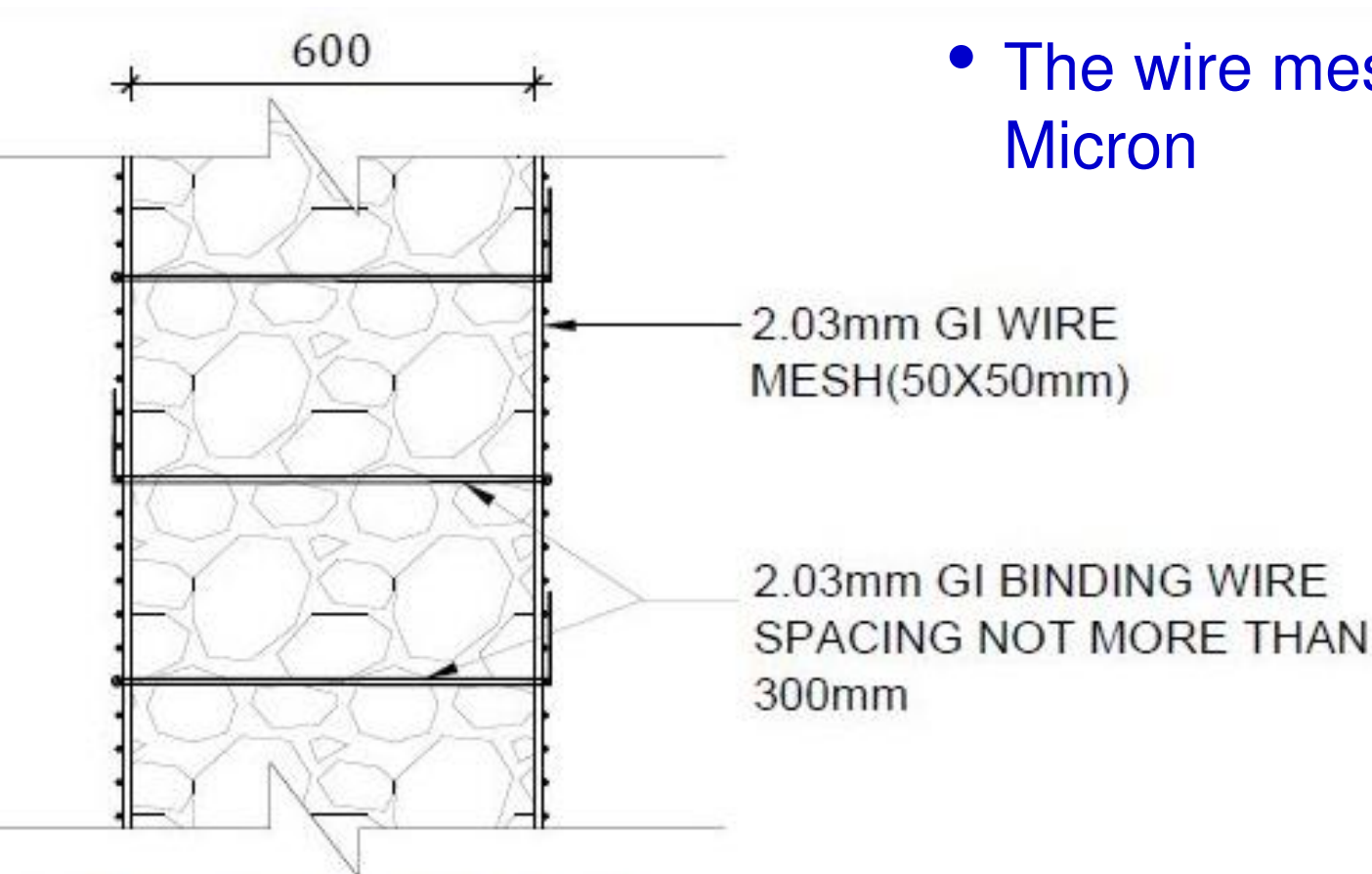
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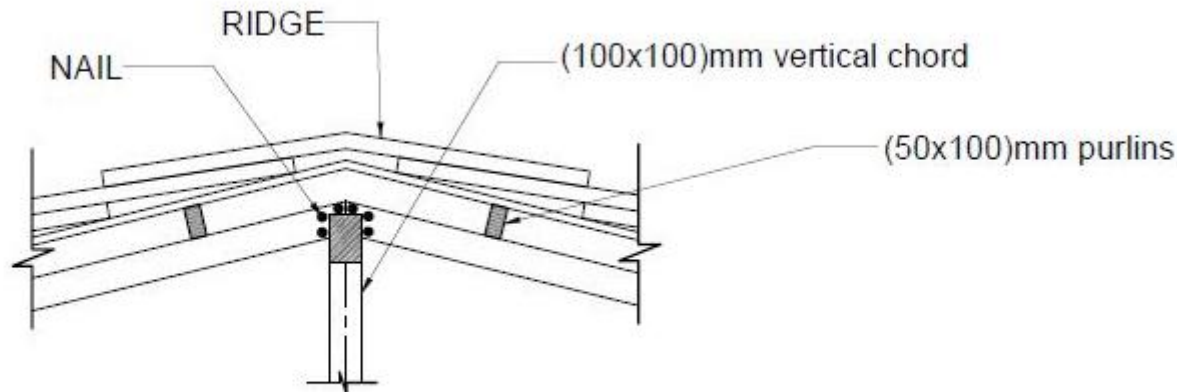


- The hot dipped galvanized welded wire Shall be used to ensure the Corrosion Protection
- The wire mesh shall not be less than 50 Micron

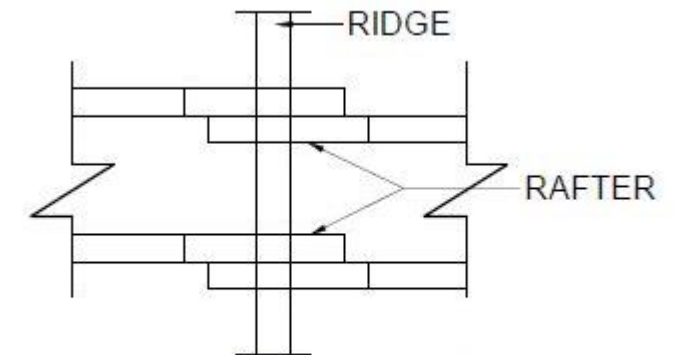


BINDING WIRE DETAIL

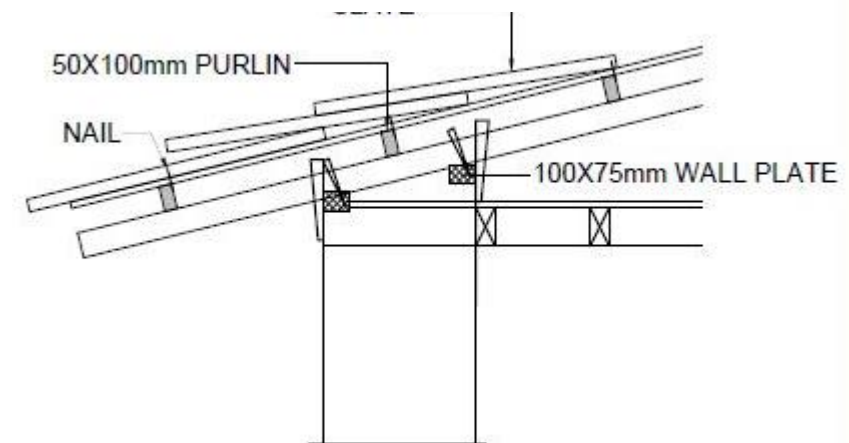
Roof Detailing



DETAIL OF RAFTER JOINT AT RIDGE

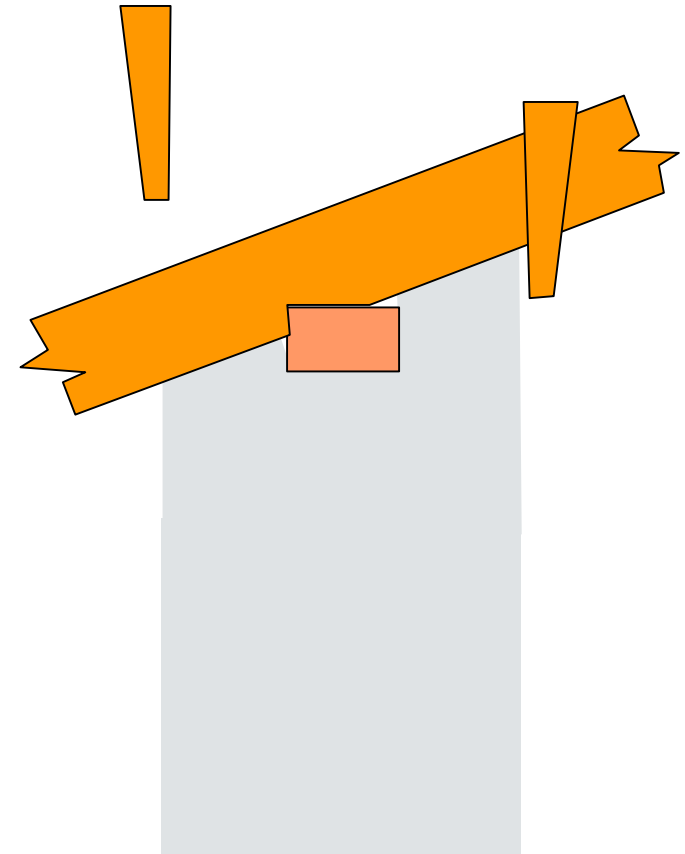
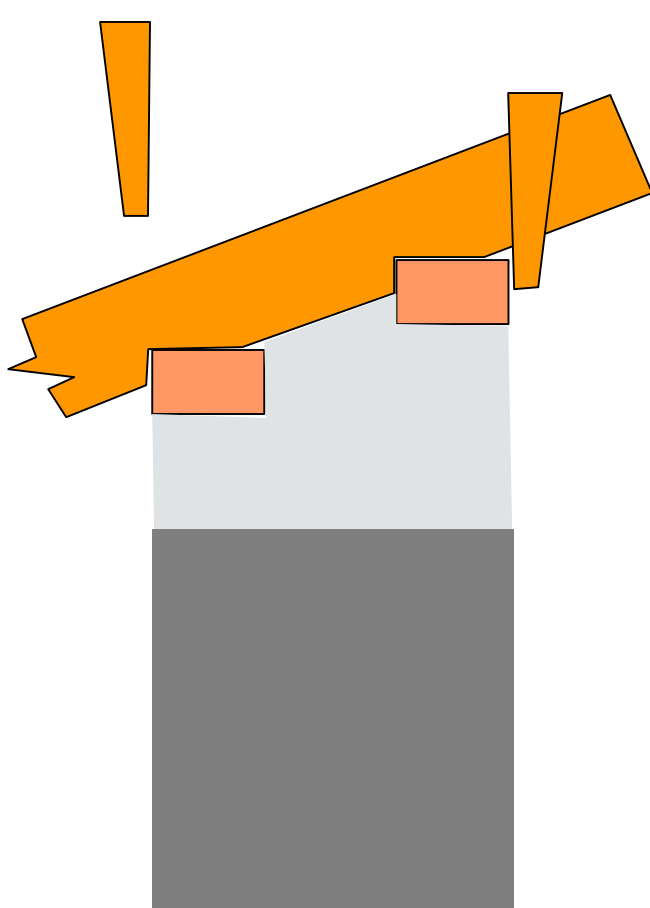


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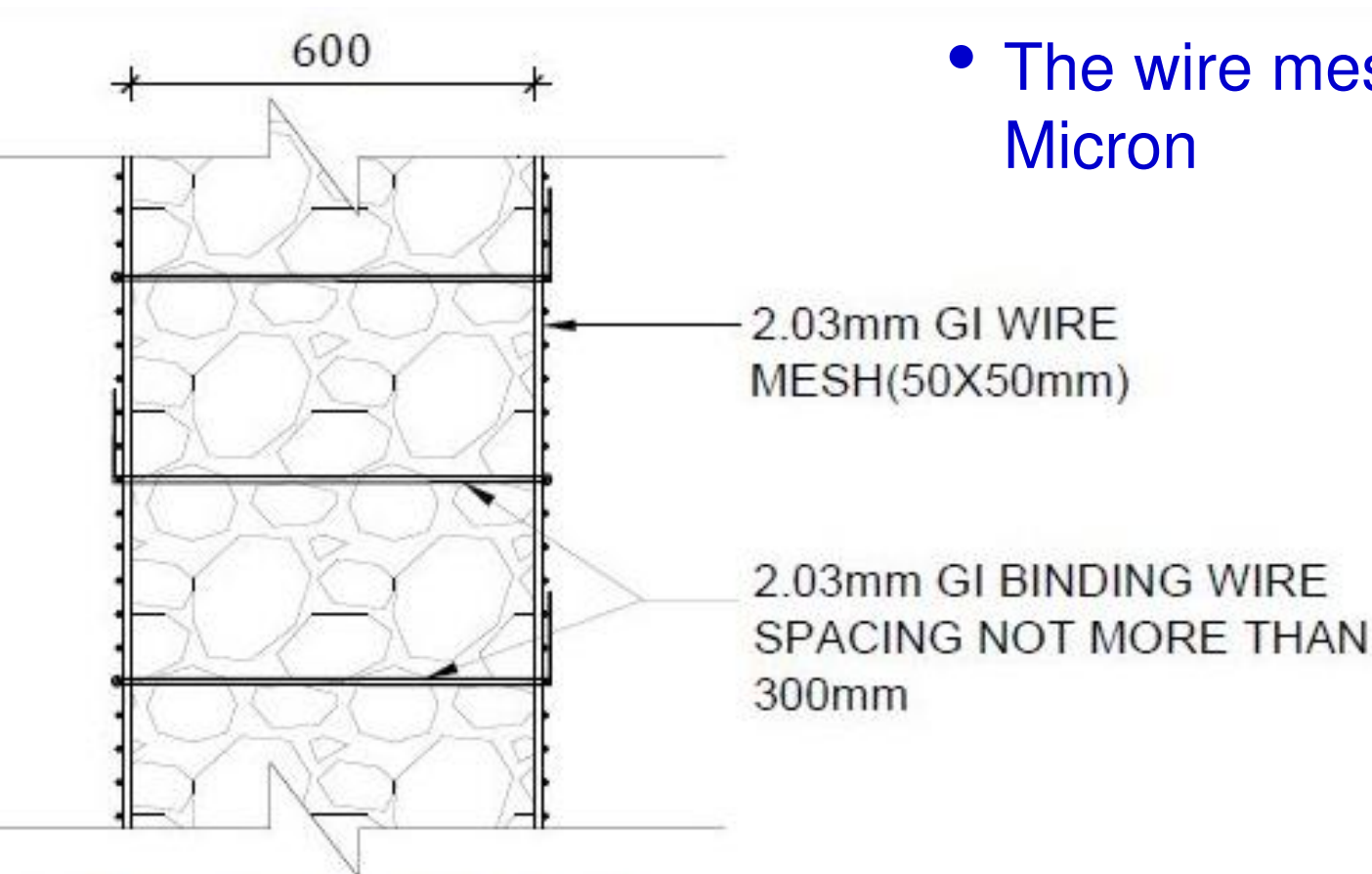
Wooden Key



Random Rubble Masonry with GI wire Mesh



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BINDING WIRE DETAIL

Even with the low compliance rate
We can
still Considered it as a good
Initiation !

Tipling, Dhading



Ramche, Rasuwa



Sertung, Dhading



Dhading



Thank You

